127. Assertion (A): The USA reemerged as India's single largest import source in the early nineties.
Reason (R): With swift political developments in the erstwhile Soviet Union, India gradually began to rely on the USA for its defence requirements. In the context of the above two statements, which one of the following is correct?
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not a correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

Ans. (c) : The USA re-emerged as India's single largest import source in the early nineties. But with swift political developments in the erstwhile Soviet Union, India did not depend on America for its defence needs.

128. The oldest monarchy in the world is that of
(a) U.K. (b) Nepal.
(c) Saudi Arabia (d) Japan

Ans. (d) : According to legend, the Imperial House of Japan was founded in 660 BCE by Japan's first Emperor, Jimmu, making it the oldest continuous hereditary monarchy in the world. Although Japan's monarch has mythological origins, the country recognizes February, 11, 660 BCE as the official date of its founding.

129. Which one of the following regions of Asia is experiencing the highest annual growth rate of population ?
(a) South Asia (b) South East Asia
(c) Central Asia (d) West Asia

Ans. (c) : The annual rate of population of the Central Asia region has been more than 2.5% during 2000-05 and 0.3% during 2005-10. So. Central Asia regions is experiencing the highest annual growth rate of population.

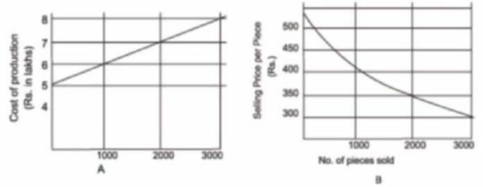
130. A major shift in the 8th FiveYear Plan from its preceding ones is
(a) the significant reduction in pubic sector outlays
(b) the concentration of pubic investment in infrastructural sectors
(c) major investment in agriculture with a view to promote exports
(d) major investment in sectors in which industrial sickness has been a chronic problem

Ans. (a) : Eighth Five Year Plant 1992 – 1997
Objectives:
Creation of employment, primary health facilities, check population growth, Drinking water & vaccination in all villages and over all human development.
Growth and diversification of agricultural activities.
Annual target Growth rate: 5.6%
Achieved growth = 6.7&
Target percapita income growth: 3.8%
Achieved Growth = 4.4%
The Eighth plan started in April, 1992. One of the Major highlight was modernization of the industries. The plan was launched with twin objectives of alleviation of poverty and unemployment.

131. 'Self sufficiency' in food. In the true sense freedom from hunger, has not been achieved in India in spite of a more than threefold rise in food grains production over 1950-1990.
Which of the following are the reasons for it ?
I. The green revolution has been restricted to small pockets of the country.
II. The cost of food is too high compared to the earnings of the poor.
III. Too much emphasis is laid on wheat and paddy compared to the coarse grains.
IV. The gains of the green revolution have largely accrued to the cash crop rather than food crops.
Select the correct answer from the codes given below.
Codes :
(a) I, II and III (b) I, II and IV
(c) I, III and IV (d) II, III and IV

Ans. (a) : There are multiple understandings of food self sufficiency that can apply at different levels of analysis. According to FAO, "The concept of food self-sufficiency is generally taken to mean the extent to which a country can satisfy its food needs from its own domestic production" (FAO, 1999). This most basic definition can apply at the level of individuals, countries, or regions.
The green revolution has been restricted to small pockets of the country in 1950-1990, the cost of food is too high compared to the earnings of the poor & too much emphasis is laid on wheat and paddy compared to the coarse grains are also a reasons for hunger.

132. Consider the following figures marked A and B :



The manufacturing cost and projected sales for a product are shown in figures A and B respectively. What is the minimum number of pieces that should be manufactured to avoid a loss ?
(a) 2000 (b) 2500
(c) 3000 (d) 3500

Ans. (a) : From the graph 'B', we know that there is cost on the construction of two thousand pieces = 7 lakh Rs.
And two thousand pieces selling price = 2000 × 350
= 7 lakh Rs.
In this way, there will be no loss on the construction of at least Rs. 2000 pieces.

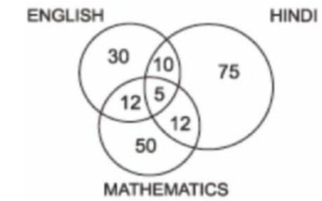
133. A man jumped at a speed of 5 metres per second from a stationary boat and the boat moved off with the speed of 0.5 metre per second. How many times is the mass of the boat greater than that of the man ?
(a) 5.5 times (b) 4.5 times
(c) 2.5 times (d) 10 times

Ans. (d) : Let the mass of the person be m and that of the boat be n, the momentum of the boat and the person before jumping = 0
Momentum of both after the person jumps = m×5 – n×0.5
From the principle of conservation of momentum,
m×5 – n×95 = 0
or n×0.5 = m×5
$\Rightarrow \frac{n}{m} = \frac{5}{0.5} = \boxed{10 \text{ times}}$

134. In the case of a Test tube baby'
(a) fertilization takes place inside the test tube
(b) development of the baby taken place inside the test tube
(c) fertilization takes place outside the mother's body
(d) unfertilized egg develops inside the test tube

Ans. (a) : The term "test-tube baby", prior to the development of in vitro fertilization technologies in the twentieth century, was used to refer to babies born as a result of artificial insemination, William, Pancoast, a physician from Philadelphia, performed the first artificial insemination that led to a successful birth in 1884, marking the birth of the first test-tube baby.
A test-tube baby is the product of a successful human reproduction that result from methods beyond sexual intercourse between a man and a woman and instead utilizes medical intervention that manipulates both the egg and sperm cells for successful fertilization.

135. Consider the diagram given below :



Five hundred candidates appeared in an examination comprising tests in English, Hindi and Mathematics. The diagram gives the number of candidates who failed in different tests. What is the percentage of candidates who failed in at least two subjects ?
(a) 7.8 (b) 0.078
(c) 6.8 (d) 1.0

Ans. (a) : Failed candidates in English & Hindi = 10%
Failed candidates in Math & English = 12%
Failed condition in Math & Hindi = 12%
Failed candidates percentage in three subjects = 5
So, failed candidates in two or more = 12 + 12 + 10 + 5
= 39
Failed candidates in minimum two subjects
$= \frac{39}{500} \times 100 = 7.8\%$

136. Sound waves travel at 300 m/s. Sound produced at a point is heard by a person after 5 seconds while the same sound is heard by another person after 6 seconds. What could be the maximum and minimum distances between the two persons?
(a) 1.8 km, 0.15 km (b) 2.2 km, 0.20 km
(c) 2.8 km, 0.25 km (d) 3.3 km 0.3 km

Ans. (d) : When both listeners in one side then the distance between them = 6×300 – 5×300 = 1800 – 1500
= 300 m or 0.3 km
When both are in different side then,
Distance = 6×300 + 5×300 = 3300 m = 3.3 km

137. The surface area of a cube is 216 sq.m. What is its volume ?
(a) 100 cu m (b) 216 cu m
(c) 480 cu m (d) 512 cu m